# Extract, then select: a modular pipeline of LLMs and rules to identify seizure frequency in epilepsy clinic letters

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***Abstract*—Seizure frequency patterns are recorded in clinical notes stored in electronic health records (EHRs). Manual chart review is required to extract this information for research, but this is expensive and error-prone. Rules-based systems struggle with the lexical variety of seizure frequency descriptions. Large language models (LLMs) are substantially better at seizure frequency extraction, but fine-tuning the model for this task requires significant data and computation. This paper proposes a modular pipeline with LLMs and rules as configurable components. Three configurations were evaluated: `Rules-only`, `LLM-only` (two prompted LLM calls) and `Hybrid` (one LLM call followed by rules). Evaluation was conducted on 1,200 synthetic epilepsy letters and compared with the benchmark previously reported for a fine-tuned LLM on a different held-out sample of the same corpus. Both `Hybrid` (Purist F1: 0.86) and `LLM-only` (0.85) exceeded the reported benchmark (0.81) without any training or fine-tuning. A modular pipeline is a viable alternative to fine-tuning, replacing scarce training data and computation with accessible prompt engineering and rules.**

***Index Terms*—information extraction, hybrid pipeline, large language models, seizure frequency, epilepsy**

## I. Introduction

Epilepsy affects c. 1% of the UK population [1] and is characterised by recurrent, unprovoked seizures [2]. Seizure frequency is a key indicator of the patient's disease burden and is commonly used in clinical care to determine the appropriate treatment and as a primary outcome in clinical trials. Despite the critical role of this information, the only place it is routinely recorded is in clinic letters. Clinic letters are unstructured narrative reports of a patient's recent visit to the epilepsy clinic, and they often combine descriptions of past diagnoses, current symptoms and future treatment. This information can be used to conduct retrospective studies or to identify patient cohorts for clinical trials, but extracting it into that structured form requires experts to conduct costly, error-prone manual chart review.

Natural language processing (NLP) has been used to automatically extract key information from clinic letters. Rules-based methods successfully extracted structured facts about medications and diagnoses, but struggled with the imprecise language and diverse forms of seizure frequency descriptions [3]. *2 to 3 seizures a week*, *frequent bouts of seizures* and *4 since her last visit* are all valid descriptions of the same fact, but designing rules to accommodate these completely and consistently is challenging and time-consuming. Rules-based systems are highly valued by clinicians because they produce interpretable results and can flexibly incorporate local domain expertise, but a different kind of tool is needed for these semantically complex tasks.

Large language models (LLMs) perform substantially better on semantically complex tasks from instructions and a few examples alone [11], [12], including seizure frequency extraction [4]. However, unlike rules-based methods, the outputs of LLMs are not constrained to the exact text they were given. This presents new challenges for the task of information extraction (IE). Past work has shown that LLMs can invent evidence even when given the specific task of estimating a patient's seizure frequency from the letter. In one case an LLM provided diagnostic advice on prescriptions instead of citing the seizure frequency [5]. This phenomenon of 'hallucination' has been documented as a common error mode for LLMs and is a significant risk when applied to clinical information at scale. Fine-tuning an LLM on the task minimises this error but requires substantial training data and computation.

Prior work has utilised rules as a light post-processing layer to repair malformed outputs from LLMs [5], but none examine a hybrid method of LLM and rules as an alternative to fine-tuning LLMs. Rules are very effective at processing information with transparent, configurable logic. LLMs are very effective at extracting rich information from semantically dense sources. Leveraging both capabilities at distinct stages can combat the concerns about LLMs being a black box that produces hallucinations, while providing greater semantic depth with far less time spent developing rules. A hybrid method can produce a more transparent, flexible system.

This task can be broken into two stages: `extract` every candidate fact, quoting its evidence and writing it in a canonical form, then `select` which candidate answers the question. A clinic letter can describe several seizure events with varying precision and on different occasions, and deciding which to keep is a clinical policy that varies between users. A system that keeps the extracted evidence separate from the selection policy can serve several such policies from one extraction, and lets users see the final answer alongside the evidence and the policy that produced it.

The primary objective of this paper is to evaluate whether using LLM and rules as configurable components in a modular pipeline is a viable alternative to a fine-tuned LLM to extract seizure frequency patterns from epilepsy letters. The secondary objective is to identify whether LLMs and rules add value at different stages of the pipeline: `extract` and `select`. If both objectives are met, the paper will contribute a new method for seizure frequency extraction and show where each component helps, to support future research on a more general form of the method.

To answer these questions, each configuration was evaluated at each stage on a set of 1,200 synthetic epilepsy clinic letters produced by Gan *et al.* [3], who used other samples of the same corpus to fine-tune their LLM. Each letter contains a single expert-annotated seizure frequency label. The task is to process the c. 400 words in each letter and extract the one fact that best describes the patient's current seizure frequency pattern. Results are compared with the benchmark reported for the fine-tuned LLM in Gan *et al.* [3]. The two papers use different held-out samples drawn from the same corpus of 15,099 synthetic letters, so this is a comparison with a previously reported benchmark rather than a head-to-head evaluation on identical letters; Section IV-A reports the uncertainty this introduces.

The rest of the paper is organised as follows. Section II reviews prior methods for seizure frequency extraction and the framing of clinical IE sub-tasks. Section III describes the dataset, the pipeline architecture, the LLM prompt and rule design, and the evaluation protocol. Section IV presents results against the prior benchmark, then by stage, by model and by model configuration. Sections V to VII discuss implications and limitations, ethical considerations, and conclusions with future work.

## II. Literature Review

The best approach to extract seizure frequency information from epilepsy clinic letters is an active area of research, with experiments using rules-based methods [4], [6], transformers-based methods [3], [5], [7], [8] and a combination of both [9], [10] all published this decade. Rules-based methods struggled with the lexical variety of seizure frequency patterns. A key challenge with both rules-based methods and BERT-based methods was their failure to generalise to letters from a different hospital. [6], [10]. LLMs were the first method to demonstrate strong generalisation to a new dataset [3], but this was achieved by fine-tuning a model on over 1,000 synthetic letters. This paper explores whether similar performance can be achieved without any additional training or fine-tuning.

Rules have been used as a form of minimal post-processing to repair malformed outputs from LLMs [5], but none of the research has explored whether a hybrid pipeline combining an LLM and rules at distinct stages can achieve superior performance than either component alone. Rules were used to normalise frequencies and dates as a separate layer in two BERT-based pipelines [9], [10], but this has not been replicated yet for modern LLMs. This paper explores whether LLM instructions and examples can be combined with complex post-processing to replace the need for fine-tuning.

Clinical IE has been defined as the automatic extraction and encoding of clinical information from text [13]: recognising a mention and normalising it into a standard form [14]. Modern LLMs do both in one step when asked for a normalised label such as *4 per month* from a raw statement such as

> Thank you for asking me to review your patient. I am concerned that her epilepsy has worsened in recent months, with a clear pattern of deterioration. She and her partner report increasing clusters of events correlating with the perimenstrual period, suggesting a catamenial component. In particular, she describes brief focal episodes with preserved awareness that tend to occur as she is settling to sleep, at the point of drifting off, now happening on most nights of the week. These events are accompanied by a rising epigastric sensation, facial pallor, and right-hand automatisms lasting 30–60 seconds, followed by fatigue and headache. In addition, she has had three generalised tonic–clonic seizures in the last six weeks, two of which occurred during the same perimenstrual window.

Fig. 1: An example clinic letter with two distinct seizure frequency patterns highlighted as potential candidates.

*about once a week*. A further task exists in many clinical extraction pipelines but goes unmentioned: selection. This was a distinct stage in prior rules-based systems [15], but LLM pipelines bundle it into the same prompt. In a clinic letter with several seizure frequency statements, producing a single label for the patient's current status requires the model both to extract the relevant statements and to select the one that best answers the question. This paper explores whether separating `extract` from `select` in a modular pipeline can deliver comparable performance while preserving more of the evidence for downstream use.

Academic papers tend to focus exclusively on benchmark accuracy while practitioners tend to value transparency, configurability and efficiency at the cost of some accuracy [15], [16]. Model interpretability is also a critical factor clinical settings [14]. Clinicians are trained to use deductive logic to draw conclusions and they expect systems to expose transparent, interpretable evidence that they can reliably use to make decisions that demand high accountability. Past work utilises evidence spans as a training input [3] for fine-tuning or as data exhaust as part of using a rule-based method [4]. This paper explores whether preserving a transparent evidence chain from exact text to final answer can make the output both more interpretable and more accurate.

## III. Methods

### A. Data

The dataset is a collection of epilepsy clinic letters which record a UK patient's consultation with an epilepsy specialist, often a neurologist or specialist nurse [3] The NHS requires that clinic letters are sent to the patient's GP after all outpatient visits such as these. They are often the only record of a patient's diagnoses, treatments and outcomes in neurological disorders, and they are written in narrative form. An example clinic letter is shown in Figure 1.

Gan *et al.* [3] produced this dataset of synthetic letters modelled on real clinic letters from King's College London hospital. The real letters were used to create 10 template letters

TABLE I: Dataset summary

| | | Development | Test |
|---|---|---|---|
| Number of letters | | 750 | 450 |
| Seizure frequency | Frequent | 51% | 50% |
| | Infrequent | 17% | 18% |
| | Unknown | 17% | 17% |
| | Seizure free | 15% | 15% |

and they were combined with 500 short descriptions of seizure frequency patterns produced by GPT-5 to create the full set of 15,099 letters. A sample of 1,200 letters was used in this paper. Each seizure frequency statement was annotated by trained experts to create the gold-standard labels. Each letter has a single label representing the patient's current seizure frequency status.

Minimal preprocessing was applied to the raw data. Letters and annotations were stored in a single .json file. Letters are minimally cleaned to correctly parse quotes, new lines, tabs, the $\leq$ sign and nulls. The raw letters were ingested in the pipeline in full; no tokenisation, section-splitting or truncation was applied.

The dataset was split into a development set of 750 letters and a test set of 450 letters, stratified by their seizure frequency category (frequent, infrequent, unknown, seizure free) to balance the distribution across both splits. The exact breakdown can be seen in Table I.

### B. Architecture

The pipeline has two stages: `extract`, then `select`. `Extract` finds every candidate seizure frequency event in the letter, quotes the exact evidence span for each, assigns it a category and writes its label in one of the canonical forms used by the gold standard. `Select` then chooses the final answer from those candidates, either picking one or combining several. Figure 2 illustrates the pipeline with a development-set example. Either stage can be performed by rules or by an LLM.

Three configurations are evaluated (Table II). `Rules-only` uses no LLM: `regex` and if-then rules extract candidates from the whole letter and select the answer. `Hybrid` (LLM + rules) makes one LLM call, which returns a structured record of candidate events, each with a quoted evidence span, a category and a canonical label, together with a first-choice answer; rules then apply the selection policy to that record. `LLM-only` makes the same first call, then a second LLM call receives the candidate record and first choice (not the letter) and applies the same selection policy, expressed as written instructions and worked examples. The only difference between `Hybrid` and `LLM-only` is therefore who performs `select`: rules or a second LLM call.

The pipeline is a fixed sequence of calls and functions. No LLM call uses tools, chooses the next step or invokes another model, so the term *agent* is not used; each LLM call is a prompt with a fixed input and a schema-validated JSON output. Section IV-B compares one-, two- and three-call decompositions of `LLM-only` and motivates the two-call design. The supporting materials describe further configurations.

TABLE II: The three configurations: who performs each stage

| Configuration | LLM calls | Extract | Select |
|---|---|---|---|
| Rules-only | 0 | Rules | Rules |
| Hybrid (LLM + rules) | 1 | LLM (call 1) | Rules |
| LLM-only | 2 | LLM (call 1) | LLM (call 2) |

`Hybrid` and `LLM-only` share the same first call, which extracts candidates with quoted evidence and canonical labels and proposes a first choice. Call 2 receives that record, not the letter.

TABLE III: Extract and first-choice policy

| When | Do |
|---|---|
| Several current seizure types are present | Select the highest current or recent frequency. |
| The letter gives an overall current count and a breakdown by type | Select the overall count. |
| A seizure-free statement sits with other current seizure-like events | Keep seizure-free separate from unknown or last-event statements. |

TABLE IV: Seizure frequency categories

| Category | Use |
|---|---|
| `frequency_rate` | A countable or range rate |
| `cluster_frequency` | How often clusters occur |
| `seizure_free` | A seizure-free statement |
| `last_event_only` | A dated last event without a rate |
| `unknown_frequency` | Seizures discussed, no usable rate |
| `no_reference` | No usable frequency evidence |

### C. LLM Prompt Design

The most consequential component of the `Hybrid` and `LLM-only` configurations is the `extract` prompt used for the first call. It contains three key components:

1) Instructions to identify relevant seizure frequency events.
2) Examples of how the events should be represented.
3) The schema the events must be represented in.

A key aspect of the schema design is representing multiple seizure frequency patterns with exact evidence. A subsequent LLM call or rules can then review that evidence and refine the final answer. The model finds multiple candidate events and selects the best candidate, but that answer can be updated. Key instructions to choose the initial candidate are in Table III. An example of a later `select` policy that may rewrite that first pick is if the model finds the evidence *3 seizures in March, 6 in May* and the model chooses the 6 seizures in May as the most recent total, rules or a subsequent LLM call can overwrite that to include the total over the whole period i.e. `9 per 3 months`. It reuses the evidence the model found according to a specific policy.

The model is instructed to assign each event to one of the six categories described in Table IV and assign a label to each event in one of the canonical forms shown.

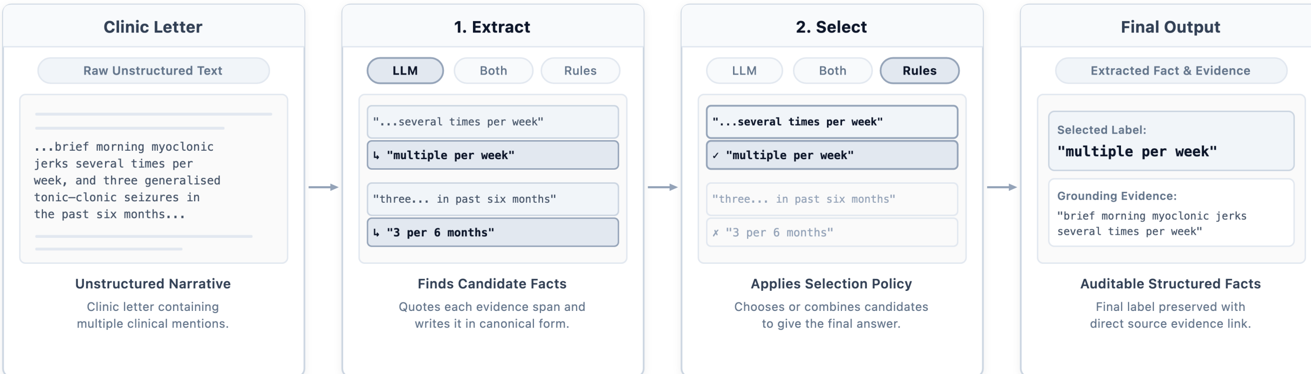


Fig. 2: A clinic letter passes through two stages (`extract`, then `select`) to produce a structured label with grounding evidence. Either stage can be performed by an LLM or by rules; the highlighted choices show the `Hybrid` configuration.

### *D. Rule Design*

Rules use `regex` to match patterns in text and if-then logic to process that text in a specific sequence. Each rule receives a pattern as an input then takes an action: `extract` a candidate from the text in canonical form, or `select` the final answer. `Rules-only` runs both stages on the entire letter. In `Hybrid`, only the `select` rules are applied, to the candidates the model returned.

The simplest `extract` rule is the countable rate: find a number (digit or word), a period word such as *per* or *times*, and a time unit, near a seizure mention. That finds $\leq$ *four per day* in the text and records a `frequency_rate` candidate labelled `4 per day`, following the gold-standard convention that flat rates and ranges are valid but single upper or lower bounds are not. In `Rules-only`, rules extract rates, clusters, seizure-free durations and dated last events across the letter, then choose between them using the policies in Table III.

In `Hybrid` the model has already returned a record of candidate events, each with quoted evidence and a canonical label, and a first pick. Rules first rewrite any label the model left in a non-canonical form (a step that changes the score by at most 0.01 and is detailed in the supporting materials), then apply the selection policies to that record. If the model finds both *seizure free for eight months* and *two seizures in the past fortnight* but chooses the seizure-free duration, a `select` rule rejects seizure-free while other current seizure-like events remain. For seizure diaries, rules apply date and numeric logic to total the events over the stated period, e.g. *May x0, Jun x0, Jul x1, Aug x0, Sep x1, Oct x2* becomes `4 per 6 months`. Rules therefore refine the evidence the LLM found with transparent, configurable policies.

### *E. Pipeline Optimisation*

The development set was used to optimise the two key elements of the pipeline: the LLM prompt instructions and the rules. The focus of the optimisation process was to improve accuracy while keeping the instructions and rules simple and generalisable. Qualitative error analysis was conducted to diagnose the cause of poor performance and identify common error modes. Hypotheses were developed and tested to tune the prompt and rules iteratively. Improving a rule or instruction for one class of error often harmed another, because the annotation policies interact. Separating `extract` from `select` kept the policy on which facts to extract apart from the policy on which to choose, and made it possible to score the pipeline after each stage and locate bottlenecks. The supporting materials give a fuller account of this process.

### *F. Evaluation*

Every paper reporting seizure frequency extraction this decade used F1 as its primary metric, but they differ in what counts as a correct frequency label. It is not obvious whether *2 per month*, *3 per month* and *once a week* should all be accepted when the gold-standard label is *multiple per month*. To allow comparison with Gan *et al.* [3], this paper adopts their two evaluation settings: a fine-grained `purist` setting and a coarser `pragmatic` setting. If the gold label is *twice a week* and the prediction is *once a week*, that is a correct `pragmatic` answer, since both fall in the `frequent` category, but an incorrect `purist` answer. Table VI lists the categories in each setting.

Micro-F1 is reported for both settings, because the categories are badly imbalanced (some have over 100 letters, some fewer than 10; Table VI) and micro-F1 gives each letter equal weight. Each letter has exactly one gold label and one prediction, so micro-F1 here equals the share of letters answered correctly. The more precise `purist` F1 is the primary metric.

The pipeline is evaluated on a held-out test set of 450 letters that were never used to tune prompts, rules or any other aspect of the pipeline. Results are compared against the evaluation set labelled Synthetic (1,166) in Gan *et al.* [3], a different held-out sample from the same synthetic corpus, distinct from their fine-tuning data.

The primary model is Gemini 3.7 Flash. It is the only model evaluated in the `LLM-only` configuration and the only model used for temperature and thinking-level ablations. Six models

TABLE V: Comparison with previously reported results

| Author | Year | Method | Purist F1 | Pragmatic |
|---|---|---|---|---|
| Holgate et al. | 2024 | Prompted LLM | - | 0.73 |
| Holgate et al. | 2025 | Fine-tuned LLM | 0.68 | 0.81 |
| Gan et al. | 2026 | Fine-tuned LLM | 0.81 | 0.85 |
| This paper | 2026 | Hybrid (LLM + rules) | 0.86 | 0.88 |

Each row was evaluated on a different held-out sample; only Gan et al. and this paper share a corpus and metric definitions.

were evaluated in the `Hybrid` configuration (Section IV-C). Elsewhere, `Hybrid` results are from Gemini.

## IV. Results

### A. Benchmark Performance

The primary objective of this paper is to evaluate whether a modular pipeline combining an LLM and rules is a viable alternative to a fine-tuned LLM for extracting seizure frequency patterns from epilepsy letters. Results are compared with the Synthetic (1,166) evaluation reported by Gan *et al.* [3]. The two evaluations use identical metric definitions and letters drawn from the same synthetic corpus, but different held-out samples. The comparison is therefore with a previously reported benchmark, not a paired head-to-head evaluation on the same letters.

`Hybrid` achieves a `purist` F1 of 0.86 (387/450), above the reported benchmark of 0.81 for the fine-tuned LLM. `LLM-only` also exceeds it (0.85; 383/450). Because micro-F1 equals per-letter accuracy here, an approximate two-proportion comparison gives a difference of +0.05 for `Hybrid` (95% CI +0.01 to +0.09, $p = 0.01$) and +0.04 for `LLM-only` (95% CI 0.00 to +0.08, $p = 0.04$). These intervals assume independent samples from the same distribution and are indicative rather than a controlled comparison. The first LLM call alone, scored on its first-choice answer, reaches 0.79, close to the benchmark. Rules applied to that record add +0.07 in `Hybrid`; the second LLM call adds +0.06 in `LLM-only`.

The pipeline is particularly effective at extracting high-frequency events. Gan *et al.* report `unknown` as their highest scoring category (0.89), with `daily` (0.70) and `more than weekly` (0.68) much lower. In this pipeline, `daily` (0.92) and `more than weekly` (0.91) are among the best performing categories, well above `unknown` (0.74). The full breakdown across `purist` and `pragmatic` categories is in Table VI.

### B. Performance by Stage

Table VII reports `purist` F1 at two stops in each configuration: the initial answer after `extract`, and the final answer after `select`. Each stop scores the answer the pipeline would submit if it stopped there. The initial answer is the extract stage's own first choice, so its score measures the quality of that proposal, not candidate identification alone; an isolated measure such as candidate recall is left to future work. The difference between the columns is what `select` adds to the answer already proposed.

TABLE VI: Hybrid: Purist and Pragmatic categories (test set, $n = 450$)

| Purist | *n* | Purist F1 | Pragmatic F1 |
|---|---|---|---|
| Daily | 41 | 0.92 | Frequent: 0.95 |
| More than weekly, less than daily | 123 | 0.91 | |
| Once a week | 5 | 0.89 | |
| More than monthly, less than weekly | 58 | 0.90 | |
| Once a month | 18 | 0.94 | Infrequent: 0.82 |
| More than every 6 months, less than monthly | 55 | 0.81 | |
| Once every 6 months | 1 | 0.00 | |
| Less than once every 6 months | 6 | 0.60 | |
| Unknown | 76 | 0.74 | Unknown: 0.74 |
| Seizure free | 67 | 0.90 | Seizure free: 0.90 |

Rows with $n$<10 shown in grey.

TABLE VII: Purist F1 after each stage, by configuration

| Configuration | Initial answer (after extract) | Final answer (after select) |
|---|---|---|
| Rules-only | 0.63 | 0.72 |
| Hybrid (LLM + rules) | 0.79 | 0.86 |
| LLM-only | 0.79 | 0.85 |

Each column scores the answer the pipeline would submit at that stop. `Hybrid` and `LLM-only` share the same first call.

The initial answer from a single LLM call (0.79) is already above the complete `Rules-only` pipeline (0.72). This single call is equivalent to the entire pipeline in most prior LLM studies, which ask the model to extract and select in one prompt. Both `Hybrid` and `LLM-only` then improve on that shared initial answer at `select` (+0.07 and +0.06). This can mean choosing a different candidate, such as a frequency candidate when the model chose `unknown`, or calculating a new label from the selected evidence, such as totalling the events in a monthly seizure diary. Applying a selection policy to the evidence already extracted is where the pipeline adds value over a single prompt, and it can be performed equally well by rules or by a second LLM call.

Table VIII motivates the two-call design of `LLM-only`. Asking one prompt to extract and select together scores 0.79, the same as the initial answer. Separating `select` into a second call raises this to 0.85. Splitting the work across three calls, either by dividing extraction into two prompts or by adding a call that rewrites the extracted labels before selection, returns to 0.79: the extra call overwrote answers that were already correct. More calls are not always better; one call to extract and one to select is the smallest decomposition that helps.

TABLE VIII: LLM-only decompositions by number of calls

| Decomposition | Calls | Purist F1 |
|---|---|---|
| Extract and select in one prompt | 1 | 0.79 |
| Extract / select | 2 | 0.85 |
| Extract in two prompts / select | 3 | 0.79 |
| Extract / rewrite labels / select | 3 | 0.79 |

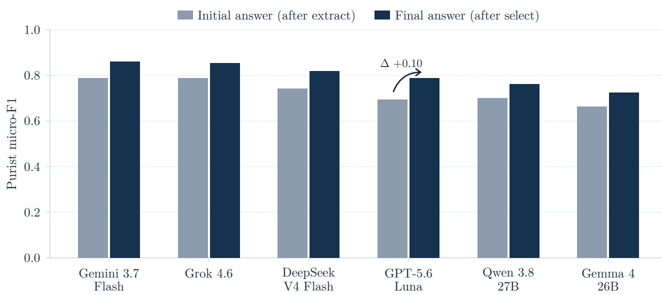


Fig. 3: `Hybrid` Purist F1 for six models: the LLM's initial answer after `extract` and the final answer after rule-based `select`. Every model gains at `select`; Luna gains most (0.69 to 0.79).


Fig. 4: `Hybrid` confusion matrix on the `pragmatic` categories (test set). Most errors are incorrect predictions of `unknown`.

## C. Model Comparison

`Hybrid` shows consistent, incremental gains on each of the six models evaluated (Figure 3). The average gain from the initial answer to the final answer was +0.07, with Luna showing the largest gain of +0.10. `Purist` F1 closely tracks general medical capability as measured by Artificial Analysis' Healthcare and Medical Index. The models cluster into three groups: strong (Gemini 0.86, Grok 0.85), medium (DeepSeek 0.82, Luna 0.79) and weak (Qwen 0.76, Gemma 0.72), which largely reflect model size and cost.

Table V also lists the two earlier papers that used the same `pragmatic` metric on different data. Holgate *et al.* [8] ran Llama 2 on local hospital servers with eight Nvidia V100 GPUs and reached a `pragmatic` F1 of 0.73 without fine-tuning. In `Hybrid`, Qwen reached a `pragmatic` F1 of 0.80 running locally on a laptop with a single GPU, and DeepSeek, which is small enough for many hospital servers to host, exceeded the benchmark reported for the model fine-tuned on 1,797 synthetic letters. Hospitals can therefore process patient data on their own hardware and still extract this information reliably.

## D. Model Configuration

Raising the reasoning effort level did not improve performance. Gemini's `purist`-F1 score was 0.86 at low and 0.84 at high; a paired difference of +0.02, 95% CI -0.01 to 0.04; p = 0.31. Extra reasoning adds cost and latency without improving extraction.

The temperature setting also had no significant effect on the results. The F1 score was 0.86 on a temperature of 0 versus 0.84 at a temperature of 1 (paired difference +0.02, 95% CI -0.01 to 0.04; p = 0.20). A temperature of 0 was an appropriate default across models and Luna at a temperature of 1 is a valid comparison.

The task is tightly defined with explicit schema and output requirements, which limits the expected variability by design. This partly explains why neither setting had a significant effect on results and it indicates the robustness of the method.

## E. Error Analysis

The worst performing category was `unknown` with an F1 score of 0.74. 69% of the errors (37/54) are incorrect selections of `unknown`. The full error distribution is shown in the confusion matrix (Figure 4).

The three most common errors are `infrequent → unknown` (16), `frequent → unknown` (13), and `seizure free → unknown` (8). The errors reflect a cautious approach when the evidence is ambiguous, for example the model recognised the phrase 'these episodes crop up most shifts' as the primary seizure frequency description but refused to make an assumption about how many shifts the person works. The gold-standard label was `multiple per week` but the model chose `unknown`. This cautious assessment is justifiable, and because the exact evidence is preserved, downstream users can see the proposed answer and draw their own conclusion from the quoted span.

## F. Method Generalisability

To check whether the two-stage design transfers beyond a single-label task, `Hybrid` was also configured for the ExECTv2 epilepsy annotation dataset [4], which asks for a complete inventory of diagnoses, prescriptions and investigations per letter in a nested schema with a closed vocabulary. On its held-out letters `Hybrid` reached F1 of 0.84 for diagnosis, 0.93 for prescriptions and 0.93 for investigations, comparable to the rules-based system reported on that dataset [4]. This is a preliminary result on a separate dataset with its own scorer, reported only to show that the same `extract` and `select` stages can be reconfigured for a different task; it will be described fully in a separate paper.

### G. Development Environment

Experiments using local language models took place on a Dell XPS 16 9640 running Windows 11, with an Intel Core Ultra 9 185H, 32 GB RAM, and an NVIDIA GeForce RTX 4070 Laptop GPU (8 GB VRAM). Ollama v0.32.15 was used to serve the models. Two models were run on local hardware (Qwen 3.8 27b, Gemma 4 26B).

Experiments using hosted APIs were run from a Mac mini M1 with 16 GB unified memory on macOS 26.5.2. API calls were made directly to the provider (OpenAI, DeepSeek) or through model routers (OpenRouter, AI Gateway) Four hosted models were used (Gemini 3.7 Flash, GPT 5.6 Luna, Grok 4.6, DeepSeek v4 Flash).

All models used a temperature of 0 except Luna, which only accepts a temperature of 1. An ablation was run on Gemini using a temperature of 1. All models used a thinking or effort level of 'low'. Ablations were run on Gemini using thinking levels of medium and high.

Python 3.14.4 was used to write the program. DSPy 3.2.1, Pydantic 2.13.4 and LiteLLM 1.87.1 were utilised as core packages. DSPy was used to co-ordinate the model calls and require structured JSON output. Pydantic was used to validate the structured output conforms to the specified schema. LiteLLM was used to deliver API calls to Ollama and the model routers.

## V. Discussion

This paper shows that a modular pipeline of LLM and rules can extract an epilepsy patient's current seizure frequency pattern as accurately as a fine-tuned LLM [3]. An LLM small enough to fit on a hospital's local servers can securely and reliably extract this information with no additional training or fine-tuning; a single LLM prompt designed to extract rich evidence followed by some rules to refine that evidence is just as accurate. This method requires far less computing power, data and expertise than training or fine-tuning an LLM, lowering barriers to adoption. Broader use of automated clinical extraction pipelines can help expand the scope of real-world evidence (RWE) as an alternative to expensive clinical trials [17].

The method is particularly effective at identifying patients with the greatest disease burden; those with the most frequent seizures. F1 scores of 0.92 for daily and 0.91 for more than weekly are around 0.2 higher than the category scores previously reported [3], although on a different sample. This could be used in a few scenarios. It could be used in clinical care to flag patients with a seizure frequency above a typical threshold for the clinician's attention. It could be used in cohort selection to identify patients to be enrolled in clinical trials for a new medication or to be recommended for surgery. It could be used to build a phenotypic profile of patients to be used in retrospective analyses to understand how these patients differ from the average. Finally, it could be used as an outcome variable in predictive models to identify patients likely to experience frequent seizures and recommend an early intervention [13].

The latest generation of LLMs are now capable enough to reliably populate rich schemas of facts and relations. Past work in this area has typically tasked the LLM with producing a single answer in a fully normalised form [3]. This fails to leverage the rich semantic depth both in clinical narratives and within the model's capabilities. Clinicians value unstructured narrative because it is more expressive; it allows them to convey their reasoning, uncertainty and vague impressions. These nuanced observations are not only more honest representations, they are more accurate and reliable [18]. LLMs are capable of preserving and reasoning over those nuances, and preserving this richer evidence chain can help build a more trustworthy, transparent system. In a clinical setting, a more explainable system is often a more useful system [14].

There are a few limitations. Synthetic data was used for all evaluations, and while past research has shown models fine-tuned on this dataset generalise effectively to real world patient data [3], the methods tested in this paper remain untested on real patient data. Past research has shown that rules tuned on one corpus can generalise poorly to other corpuses, so the gains seen from rules in this method may not generalise to other datasets [6]. As this method was applied to a narrow task of seizure frequency extraction, future studies should evaluate whether it generalises to other common tasks in clinical IE. Finally, the stage scores in this paper are cumulative stops rather than isolated stage metrics; task-specific measures such as candidate recall at `extract` would give a cleaner attribution of each stage's contribution. The fixed pipeline could also be reconfigured so that an LLM decides how to represent and `select` facts within a more general framework, which has shown promising results on clinical notes in oncology [19] and cardiology [20].

## VI. AI Ethical Considerations

This evaluation used only the synthetic clinic letters released by Gan *et al.* [3]. Those letters are modelled on real UK epilepsy letters but do not contain identifiable patient records. Hospitals prefer processing patient data on their own servers as it is highly sensitive [5]. Two of the six models were run entirely on local hardware to ensure the method transfers to privacy-maximising implementations.

LLMs can invent evidence, so careful consideration was given to the prompt instructions and schema design. The model is required to quote exact evidence spans from the letter, and every later change to the answer is made by a named rule or a recorded second call applied to that quoted evidence. A reviewer can therefore trace any final label back to the text that supports it and to the policy that selected it [14]. This transparency matters more than headline accuracy in a clinical setting, where an unexplained answer cannot be safely acted on.

The letters contain no demographic attributes, so unequal performance across patient groups cannot be tested, and the synthetic corpus may not reflect the language of every clinic. Bias can enter through the models, the prompts and the selection rules; for example, the cautious default to `unknown` (Section IV-E) would systematically under-count patients whose seizures are described vaguely. Rules make the selection policy inspectable and changeable by local clinicians, but they do not remove the need for local clinical review and fairness testing on real records before any operational use.

## VII. Conclusion

Seizure frequency is the most common outcome measure in epilepsy clinical trials and there is a wealth of this information stored in EHRs. Manual chart review by clinical experts is the most common approach to extract this information for research, but this is costly, error-prone and time consuming.

This paper proposes a modular pipeline using LLMs and rules as configurable components to extract this information. The pipeline has two stages: extracting every candidate fact with its quoted evidence in a canonical form, then selecting the final answer.

Both `Hybrid` (Purist F1: 0.86) and `LLM-only` (0.85) exceeded the benchmark previously reported for a fine-tuned LLM (0.81) on a different held-out sample of the same corpus. A single LLM call produces an initial answer (0.79) that already exceeds the complete `Rules-only` pipeline (0.72). Applying a selection policy to the evidence the LLM found adds a further +0.06 to +0.07, whether that policy is applied by rules or by a second LLM call.

Automatically extracting seizure frequency patterns from EHRs can make this critical outcome measure available for large-scale clinical research. This can be used to identify patient cohorts for future clinical trials, conduct retrospective studies or build predictive models for risk stratification. This method achieves that without requiring any data or compute for fine-tuning.

Future research should evaluate this modular pipeline on real patient data and on a broader set of clinical information extraction tasks.

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